

## OMSK, Russian Federation

WMO: 286980

Lat: 55.017N

Lon: 73.383E

Elev: 122

StdP: 99.86

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -33.1      | -30.5 | -36.7                         | 0.1 | -33.1 | -33.9 | 0.2 | -30.4 | 10.7                  | -5.9 | 9.5 | -7.9 | 1.9                   | 60   | 0.648 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.8                   | 31.0            | 18.7 | 29.0 | 18.3 | 27.2 | 17.6 | 20.6                | 27.6 | 19.7 | 26.4 | 18.9 | 25.1 | 4.1                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 13.3 | 22.9 | 17.2 | 12.5 | 22.1 | 16.5 | 11.9 | 21.7 | 60.1          | 27.6 | 57.1 | 26.5 | 54.2 | 25.1 | 24.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.3          | 8.8  | 7.8 | -35.9                      | 33.9 | 3.2                | 2.0 | -38.2                                              | 35.3 | -40.0      | 36.5 | -41.8      | 37.6 | -44.1      | 39.1 |    |    |
| (5)               |      |     | -35.9                      | 22.4 | 3.2                | 1.0 | -38.1                                              | 23.1 | -40.0      | 23.7 | -41.7      | 24.3 | -44.0      | 25.0 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.3   | -17.7 | -15.0 | -6.7  | 4.5   | 12.5  | 17.8  | 19.4  | 17.1  | 10.9  | 3.6   | -6.7  | -13.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 14.36 | 7.61  | 6.96  | 5.87  | 5.99  | 5.37  | 4.42  | 3.67  | 4.09  | 4.59  | 4.95  | 7.58  | 7.68  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3786  | 859   | 699   | 518   | 182   | 36    | 3     | 0     | 1     | 43    | 204   | 500   | 741   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 6032  | 1118  | 932   | 776   | 414   | 192   | 62    | 33    | 73    | 228   | 456   | 750   | 999   | (9)  |
| (10) |                                                                                  | CDD10.0     | 959   | 0     | 0     | 0     | 18    | 114   | 238   | 292   | 222   | 69    | 6     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 164   | 0     | 0     | 0     | 0     | 12    | 47    | 67    | 35    | 4     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 1724  | 0     | 0     | 0     | 7     | 193   | 499   | 620   | 346   | 59    | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 512   | 0     | 0     | 0     | 1     | 48    | 155   | 189   | 107   | 12    | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.5   | 3.2   | 3.2   | 3.8   | 4.1   | 4.0   | 3.4   | 3.3   | 3.2   | 3.3   | 3.5   | 3.7   | 3.5   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 402   | 21    | 15    | 16    | 23    | 32    | 54    | 66    | 55    | 31    | 33    | 32    | 26    | (15) |
| (16) |                                                                                  | PrecMax     | 636   | 48    | 42    | 49    | 64    | 106   | 136   | 180   | 144   | 90    | 90    | 85    | 94    | (16) |
| (17) |                                                                                  | PrecMin     | 258   | 2     | 1     | 3     | 3     | 5     | 6     | 1     | 12    | 3     | 5     | 3     | 1     | (17) |
| (18) |                                                                                  | PrecStd     | 82    | 11    | 8     | 10    | 14    | 21    | 31    | 38    | 27    | 19    | 19    | 16    | 15    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 0.8   | 1.2   | 7.4   | 24.0  | 30.7  | 32.8  | 33.9  | 33.2  | 28.2  | 20.0  | 7.9   | 1.6   | (19) |
| (20) |                                                                                  |             | MCWB  | 0.2   | 0.2   | 3.9   | 13.3  | 16.5  | 19.2  | 20.4  | 19.5  | 15.5  | 11.3  | 6.1   | 0.6   | (20) |
| (21) |                                                                                  | 2%          | DB    | -1.9  | -0.3  | 4.1   | 20.1  | 27.8  | 30.2  | 30.8  | 29.2  | 24.8  | 16.8  | 5.8   | 0.2   | (21) |
| (22) |                                                                                  |             | MCWB  | -2.5  | -1.2  | 1.8   | 11.1  | 15.6  | 18.3  | 19.6  | 18.7  | 14.6  | 10.0  | 4.4   | -0.4  | (22) |
| (23) |                                                                                  | 5%          | DB    | -3.8  | -2.2  | 2.4   | 17.3  | 25.4  | 28.4  | 28.8  | 27.0  | 21.4  | 13.9  | 3.9   | -1.2  | (23) |
| (24) |                                                                                  |             | MCWB  | -4.5  | -2.9  | 0.8   | 10.1  | 14.5  | 18.0  | 18.9  | 18.1  | 13.5  | 8.6   | 2.6   | -1.9  | (24) |
| (25) |                                                                                  | 10%         | DB    | -6.0  | -4.5  | 1.2   | 14.2  | 22.8  | 26.2  | 26.9  | 24.9  | 18.9  | 11.1  | 1.9   | -3.5  | (25) |
| (26) |                                                                                  |             | MCWB  | -6.6  | -5.2  | -0.1  | 8.5   | 13.3  | 17.2  | 18.6  | 17.3  | 12.2  | 7.3   | 0.8   | -4.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -0.1  | 0.4   | 4.1   | 14.2  | 18.5  | 21.5  | 22.1  | 21.1  | 17.2  | 12.6  | 6.3   | 0.7   | (27) |
| (28) |                                                                                  |             | MCDWB | 0.4   | 1.0   | 6.5   | 21.5  | 27.9  | 29.1  | 29.1  | 28.6  | 25.1  | 17.8  | 7.1   | 1.1   | (28) |
| (29) |                                                                                  | 2%          | WB    | -2.5  | -1.3  | 2.1   | 12.1  | 16.5  | 20.0  | 20.9  | 19.9  | 15.4  | 10.7  | 4.4   | -0.4  | (29) |
| (30) |                                                                                  |             | MCDWB | -1.9  | -0.6  | 3.6   | 18.3  | 25.4  | 27.3  | 27.6  | 26.7  | 22.1  | 15.2  | 5.5   | 0.2   | (30) |
| (31) |                                                                                  | 5%          | WB    | -4.4  | -3.1  | 1.1   | 10.5  | 15.2  | 19.0  | 20.0  | 18.9  | 14.1  | 9.2   | 2.5   | -2.0  | (31) |
| (32) |                                                                                  |             | MCDWB | -3.8  | -2.4  | 2.2   | 16.7  | 23.5  | 25.8  | 26.7  | 24.9  | 20.5  | 13.1  | 3.6   | -1.4  | (32) |
| (33) |                                                                                  | 10%         | WB    | -6.6  | -5.3  | -0.1  | 8.8   | 13.9  | 18.0  | 19.2  | 18.0  | 12.7  | 7.5   | 0.8   | -4.2  | (33) |
| (34) |                                                                                  |             | MCDWB | -6.0  | -4.6  | 1.1   | 13.9  | 21.4  | 24.5  | 25.4  | 23.3  | 18.2  | 10.5  | 1.5   | -3.6  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.3   | 9.1   | 9.0   | 10.3  | 12.7  | 11.5  | 10.8  | 10.9  | 10.7  | 8.4   | 6.6   | 7.8   | (35) |
| (36) |                                                                                  |             | MCDBR | 8.9   | 7.7   | 8.0   | 15.1  | 16.4  | 14.6  | 14.1  | 14.8  | 15.3  | 13.5  | 6.7   | 7.7   | (36) |
| (37) |                                                                                  |             | MCWBR | 8.8   | 7.3   | 6.4   | 8.5   | 8.1   | 6.3   | 5.7   | 6.5   | 7.3   | 8.0   | 5.9   | 7.7   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 8.9   | 7.3   | 7.3   | 14.2  | 15.1  | 12.8  | 12.2  | 12.5  | 14.1  | 12.0  | 6.5   | 7.9   | (38) |
| (39) |                                                                                  |             | MCWBR | 8.8   | 7.1   | 6.2   | 8.5   | 8.4   | 6.2   | 5.7   | 6.1   | 7.2   | 7.7   | 5.9   | 7.8   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.284 | 0.316 | 0.375 | 0.406 | 0.402 | 0.409 | 0.434 | 0.432 | 0.370 | 0.354 | 0.323 | 0.289 | (40)  |      |
| (41) |                                                                                  | taud        | 2.044 | 2.015 | 1.931 | 2.128 | 2.229 | 2.289 | 2.214 | 2.186 | 2.323 | 2.237 | 2.101 | 2.007 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 610   | 729   | 765   | 810   | 840   | 838   | 805   | 778   | 785   | 690   | 552   | 496   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 68    | 102   | 141   | 135   | 129   | 123   | 130   | 126   | 96    | 82    | 65    | 57    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.97  | 2.05  | 3.62  | 4.65  | 5.61  | 5.99  | 5.51  | 4.45  | 3.14  | 1.63  | 0.79  | 0.63  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.07  | 0.09  | 0.28  | 0.37  | 0.48  | 0.36  | 0.40  | 0.35  | 0.26  | 0.20  | 0.12  | 0.07  | (45)  |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |
| (47) | Regional (11 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |

Nomenclature: See separate page